CSC 3320: System-Level Programming

Chapter 2: UNIX Utilities for Non-programmers

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Note: These slides are updated based on original notes from Raj Sunderraman, Michael Weeks, and Yuan Long

Shells

Popular shells:

- Bourne Shell
- Korn Shell
- C Shell
- Bash (Bourne Again Shell)
- Z Shell (Mac 10.15 and after)

- All have common core functionality; some differences.
- Each shell has its own programming language. (Shell programming).

Running Utilities

To run a utility, simply type the name of the utility after the prompt, and press ***Enter*** key

- Some utilities: *date*, *man*, *clear*, *stty*, *passwd*
  - Utility: date [yymmddhhmm[.ss]]
  - Utility: clear
  - Utility: man [-s section] word (specifies the section of the manual to search for the given word)

    man -k keyword (to search for manual pages that contain the specified **keyword**)
- To logout, enter CTRL-D (or *exit*)

Special Characters

stty in Linux is a utility used to modify and print terminal line settings.

- list meta characters with ***stty -a*** command

```
[mrahman21@gsuad.gsu.edu@snowball ~]$ stty -a
speed 9600 baud; rows 40; columns 178; line = 0;
intr = ^C; quit = ^\; erase = ^?; kill = ^U; eof = ^D; eol = M-^?; eol2 = M-^?; swtch = <undef>;
start = ^Q; stop = ^S; susp = ^Z; rprint = ^R; werase = ^W; lnext = ^V; flush = ^O;
min = 1; time = 0;
-parenb -parodd -cmspar cs8 -hupcl -cstopb cread -clocal -crtscts
-ignbrk -brkint -ignpar -parmrk -inpck -istrip -inlcr -igncr icrnl ixon -ixoff -iuclc ixany
imaxbel iutf8
opost -olcuc -ocrnl onlcr -onocr -onlret -ofill -ofdel nl0 cr0 tab0 bs0 vt0 ff0
isig icanon iexten echo echoe -echok -echonl -noflsh -xcase -tostop -echoprt echoctl echoke
```

^ means Control

- ^C means hold down control key (CTRL) and press the C key

Special Characters

Backspace

erase: CTRL-H — erase character before cursor

- werase: CTRL-W — erase word before cursor
- kill: CTRL-U — erase entire line
- rprnt: CTRL-R — reprint line
- To test these:
  - type some input like “one two three four” at the terminal prompt
  - move cursor around
  - try the above characters

Special Characters

rprnt: CTRL-R
Reverse interactive search starts.
User partially types the command and the full command matching the pattern will appear

```
[mrahman21@gsuad.gsu.edu@snowball ~]$ echo "this is that"
this is that
[mrahman21@gsuad.gsu.edu@snowball ~]$ echo "this is that"
this is that
(reverse-i-search)`e': echo "this is that"
```

Special Characters

intr: CTRL-C interrupt running program

- susp: CTRL-Z suspend running program

stop: CTRL-S/CTRL-Q stop printing to screen

- eof: CTRL-D give program end of file
- To test these:
  - Try a command like ***find / (e.g.,*** find . -name "*.txt"***)***
  - For ***eof***, try ***cat > testfile*** then **CTRL-D** on new line

fg command will resume the most recently stopped (suspended) job in the foreground

Some common UNIX utilities

Utilities	Feature
ls	List files
cat ,more, page, head, tail	Show file contents
mkdir, rmdir,cd, pwd	Working with directories
mv, cp, rm	Working with files
wc	Word count
file, groups, chgrp, chmod	About file permission
vi, emacs	Text editor

Pathnames

The hierarchy from a starting directory to a target file or directory.

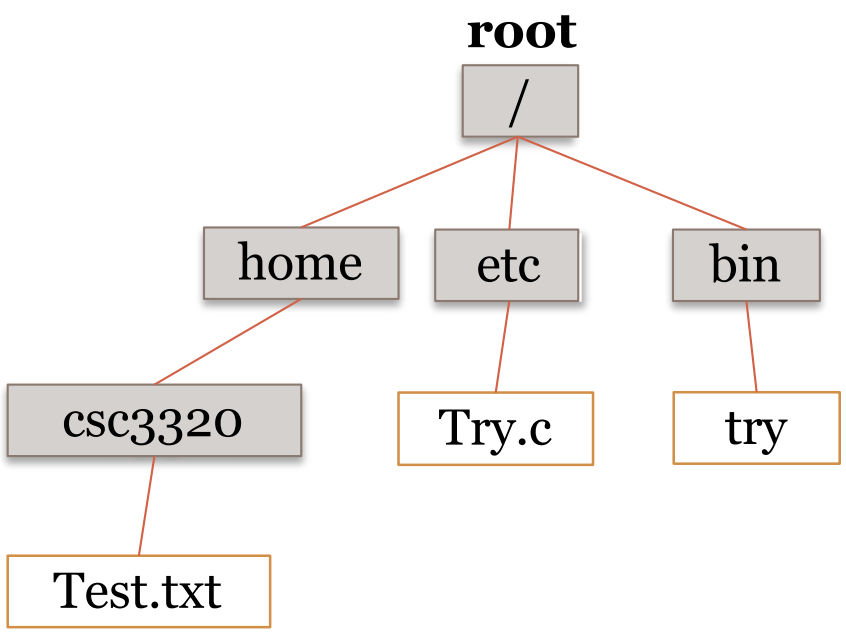
- E.g. /home/myFile , ../myFile, ./myFile

Absolute pathname : relative to the root directory

Relative pathname : relative to its current working directory

/	refers to root directory
.	refers to current directory (example: ./a.out)
..	refers to parent directory (example: ../p2.cpp)

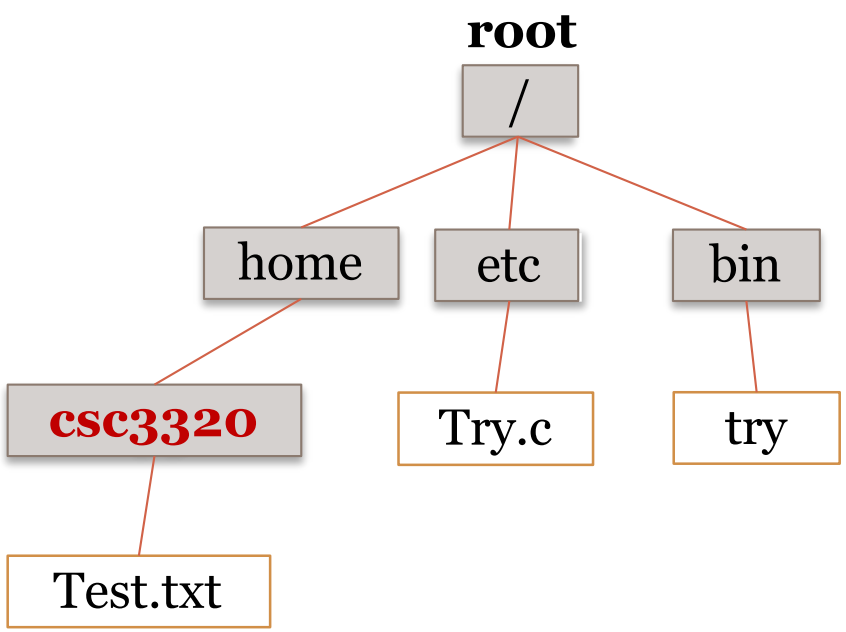
Pathnames



File	Absolute Path
Test.txt	/home/csc3320/Test.txt
Try.c	/etc/Try.c
try	/bin/try

/	refers to root directory
.	refers to current directory (example: ./a.out)
..	refers to parent directory (example: ../p2.cpp)

Pathnames



File	Relative Path
Test.txt	Test.txt or ./Test.txt
Try.c	../../etc/Try.c
try	../../bin/try

Assume csc3320 as the current working directory.

/	refers to root directory
.	refers to current directory (example: ./a.out)
..	refers to parent directory (example: ../p2.cpp)

Example of *cat*

```
$ cat > letter
Hi Mom,

Please send me money!

David
^D
$ ls -l letter
-rw-r--r--   1 raj      other        38 May 28 11:20
letter
```

Special Character

List Files (*ls*)

$ ***ls*** *-algFsdR* ***<file-spec-list>***

- options
  - *a*: hidden files
  - *l*: **long listing**
  - *g*: group
  - *F*: put character after file name indicating executable *, link @, directory /, socket =
  - *s*: number of disk blocks
  - *d*: dir details not contents
  - *R*: recursive listing

- The input can be a set of elements, separated by space
- **File-spec** is the pathname for directory or file

Show File Contents (*cat*, *more*)

$ ***cat*** ***<file-spec-list>***

- list contents of file(s) on screen without pause

$ ***more*** ***<file-spec-list>***

- same as *cat*; pauses after each screen
- **space bar** takes you to next screen
- **q** quit
- **enter** shows next line
- **h** help key for more commands

Show File Contents

$ ***page*** ***<file-spec-list>***

- same as *more*; clears screen before each page
- quicker

$ ***head*** ***–n*** ***<file-spec-list>***

- display *n* lines from front of file

$ ***tail*** ***–n*** ***<file-spec-list>***

- display *n* lines from end of file

Working with Directories

$ ***mkdir*** ***<dir-list>***

- **m**a**k**e a new **dir**ectory

$ ***cd*** ***<dir>***

- built-in to shell
- **c**hanges shell to a different **d**irectory

$ ***pwd***

- **p**rint **w**orking **d**irectory

Rename Files (*mv*)

Rename files (simply change labels in the file hierarchy)

$ ***mv*** -***i*** ***<old>*** ***<new>***

- ***i*** for inquire
- (prompt if <*new*> already exists)

$ ***mv*** -***i*** ***<file-spec>*** ***<dir>***

$ ***mv*** -***i*** ***<dir>*** ***<dir>***

Move file to a new directory

Rename a directory

Copy Files (*cp*)

$ ***cp*** ***-i*** ***<old>*** ***<new>***

- ***i*** for inquire
- (prompt if <*new*> already exists)

$ ***cp*** ***-i*** ***<file-spec>*** ***<dir>***

$ ***cp*** ***-ir*** ***<dir>*** ***<dir>***

- ***r*** for recursive

Removing Files (*rm, rmdir*)

remove (delete) file

- $ ***rm*** ***-fir*** ***<file-spec-list>***
  - **f**: force; inhibits all prompts/messages
  - **i**: inquire
  - **r**: recursive
- **Be careful!** This is a permanent deletion!
- remove (delete) **dir**ectory
- $ ***rmdir*** ***<dir-list>***
  - rmdir: failed to remove 'CSCSpring2025/': Directory not empty

Word Count (*wc*)

word count

- $ ***wc*** ***-clw*** ***<file-spec>***
  - **c** counts the bytes
  - **l** counts the newline characters
  - **w** prints the word counts

Try this: find . -type f -name "*.pdf" | wc -l

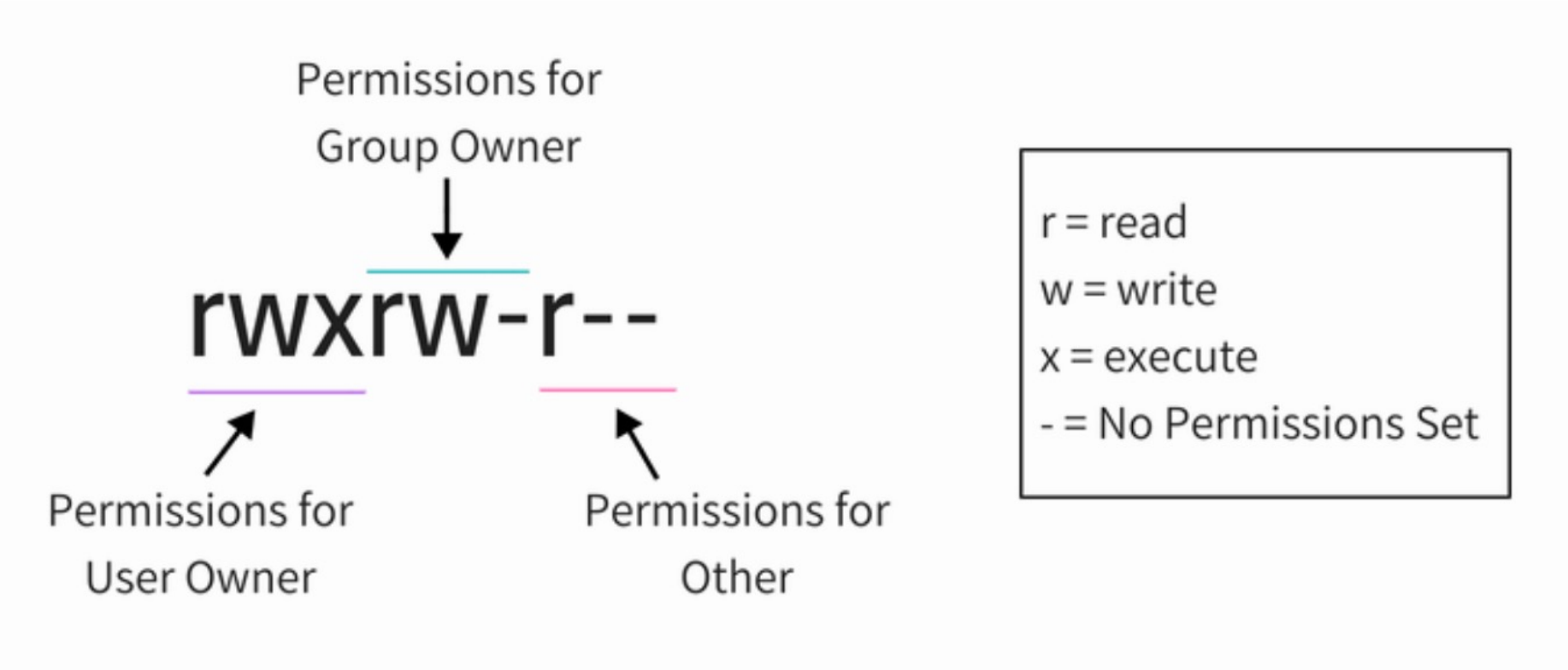
File Attributes

```
$ ls  -lsF  test.java
4 -rw-r--r--. 1 GSUAD\ylong4 GSUAD\domain^users 247 Aug 22 13:36 test.java
```

- 4 : number of physical blocks
- -rw-r--r-- : file type (first char), permissions
  - file type (**-** regular, **d** dir, **b** buffered file disk drive, **c** unbuffered file terminal, **l** link, **p** pipe, **s** socket)
- 1 : hard link count
- GSUAD\ylong4 : file owner
- GSUAD\domain^users : file's group
- 247 : file size in bytes
- Aug 22 13:36 : file modification date
- test.java : file name

How to interpret permissions?

```
(base) Mds-MacBook-Pro:scripts mdmahfuzurrahman$ ls -l test.c
-rw-r--r--  1 mdmahfuzurrahman  staff  131 Sep  1 23:14 test.c
```



File Type (file)

$ ***file*** ***<file-spec-list>***

Ascertains the type of file (ascii/binary etc.)

```
(base) Mds-MacBook-Pro:scripts mdmahfuzurrahman$ file array.c
array.c: c program text, ASCII text
(base) Mds-MacBook-Pro:scripts mdmahfuzurrahman$ file *
array:                 Mach-O 64-bit executable x86_64
array.c:               c program text, ASCII text
class_example.sh:      POSIX shell script text executable, ASCII text
example.sh:            POSIX shell script text executable, UTF-8 Unicode text
my_example1.sh:        POSIX shell script text executable, UTF-8 Unicode text
regex_exmple.sh:       POSIX shell script text executable, UTF-8 Unicode text
result_analysis.ipynb: HTML document text, ASCII text, with very long lines
test:                  Mach-O 64-bit executable x86_64
test.c:                c program text, ASCII text
test1:                 ASCII text
(base) Mds-MacBook-Pro:scripts mdmahfuzurrahman$
```

*** matches all files in current directory**

Groups (*groups,chgrp,newgrp*)

- list the groups a particular user belongs to
  - $ *groups* *<userid>*
- change the group a file belongs to
  - $ *chgrp -R <group-name> <file-spec>*
  - R stands for recursive
- create subshell with effective group id = group
  - $ *newgrp* *<group-name>*
  - your original shell waits when you run *newgrp* *<group-name>*

Groups

Example

- I belong to multiple groups

```
[mrahman21@gsuad.gsu.edu@snowball ~]$ groups
mrahman21@gsuad.gsu.edu ora_vfr_12c@gsuad.gsu.edu duo-mfa@gsuad.gsu.edu us-dropb
ox-access@gsuad.gsu.edu domain users@gsuad.gsu.edu eg-mu-coas@gsuad.gsu.edu eg-a
ff-faculty@gsuad.gsu.edu research-trends-pba@dmd.gsuad.gsu.edu
[mrahman21@gsuad.gsu.edu@snowball ~]$ date > mydatefile
[mrahman21@gsuad.gsu.edu@snowball ~]$ newgrp eg-aff-faculty@gsuad.gsu.edu
[mrahman21@gsuad.gsu.edu@snowball ~]$ date > mydatefile2
[mrahman21@gsuad.gsu.edu@snowball ~]$ ls -lg mydatefile mydatefile2
-rw-rw-r--. 1 mrahman21@gsuad.gsu.edu      29 Sep  5 02:05 mydatefile
-rw-r--r--. 1 eg-aff-faculty@gsuad.gsu.edu 29 Sep  5 02:06 mydatefile2
[mrahman21@gsuad.gsu.edu@snowball ~]$
```

id -g –n (to check which group you are currently acting)
You can terminate your current group using ^D

Change File Permissions (chmod)

change file mode; u,g,o,a r,w,x,s

- **u**ser, **g**roup, **o**ther, **a**ll
- **r**ead, **w**rite, e**x**ecute, (**s**et)
- plus (+) adds permissions
- minus (-) subtracts

-rwxr-xr--:
Owner (rwx): Can read, write, and execute.
Group (r-x): Can read and execute, but not write.
Others (r--): Can only read.

Change File Permissions

- Examples :

  Assume the file permission is `rw-r--r--`

  - $ chmod g+w <file-spec>
  - $ chmod u-rw <file-spec>
  - $ chmod u+w, g-r <file-spec>

Change parameters	Feature	File permission Result
g+w	Add group with write permission	`rw-rw-r--`
u-rw	Remove user read and write permission	`---r--r--`
u+x, g-r	Add execution permission for user, and remove read permission from group	`rwx---r--`

Change File Permissions

Assign permissions absolutely using octal number

- Example: Set the file permission of one file as **rw-r--r--**

  $chmod 644 <file-spec>

	User	Group	Others
setting	rw-	r--	r--
binary	110	100	100
octal	6	4	4

Change File Permissions

Another example with octal numbers

```
mweeks@carmaux:~$ ls -l example.txt
-rw-r--r-- 1 mweeks mweeks 51 2007-08-23 16:06
example.txt
mweeks@carmaux:~$ chmod 765 example.txt
mweeks@carmaux:~$ ls -l example.txt
-rwxrw-r-x 1 mweeks mweeks 51 2007-08-23 16:06
example.txt
```

Notice

- 7 = 111 (binary) r “on”, w “on”, x “on”
- 6 = 110 (binary) r “on”, w “on”, x “off”
- 5 = 101 (binary) r “on”, w “off”, x “on”